Cevahir Koprulu

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Research Interests

Keywords: Reinforcement learning (RL), Generalization, and Curriculum Learning

My research focuses on sample-efficient reinforcement learning (RL): what tasks an agent trains on and what signal it learns from. I designed automated curriculum generation methods for heavy-tailed task distributions, task specifications, and constrained RL. I also developed reward synthesis methods for learning from demonstrations, combining task-agnostic experience with a few demonstrations, and for offline RL, using physics-informed learned dynamics models with distance-aware uncertainty estimates. Ongoing work extends curriculum generation to composing traffic behaviors that narrow the behavior generalization gap in self-play RL, and to unlearning in large language models (LLMs) to forget with less collateral damage to retained capabilities. During internships at Waymo, Bosch Center for Artificial Intelligence (BCAI), and Honda Research Institute (HRI), I worked on training planner foundation models from unsupervised driving data, scaling curriculum learning for large-scale self-play RL in batched driving simulators, and knowledge distillation from generalist to specialist agents.

Education

University of Texas at Austin, Austin, TX, USA — M.S. 2023, PhD 2027
Electrical and Computer Engineering — GPA: 3.94/4

Bilkent University, Ankara, Turkey — B.Sc. June 2021
Electrical and Electronics Engineering — GPA: 3.73/4

Honours/Awards: (i) Bilkent University Comprehensive Scholarship (ii) Scholarship of Turkish Prime Ministry (iii) Bilkent University EEE Department High Honours (iv) Ranked 24^{th} among 2 million students in university entrance exam (v) FIRST Robotics Competition 2015: Rookie All-Star

Work Experience

Waymo — Mountain View, CA
Software Engineering Intern (all work is under NDA) || Python, SQL, JAX — June 2026 - August 2026

- Researched **learning from unsupervised driving data within Waymo's data flywheel** for onboard planner models, with a focus on undesirable driving behaviors. Surveyed state-of-the-art methods, implemented and compared a selection within the training infrastructure, and proposed an algorithm and data roadmap for future deployment.

Bosch Center for Artificial Intelligence — Sunnyvale, CA
Research Intern (all work is under NDA) || Python, PyTorch — May 2025 - August 2025

- Developed **CL4AD**, the *first* integration of curriculum learning into GPUDrive, a batched autonomous driving simulator, reducing wall clock time to achieve 99% success rate by 77%, more than a billion steps earlier than domain randomization, i.e., the default uniform scenario sampling method.

Honda Research Institute - US — Ann Arbor, MI
Research Intern (all work is under NDA) || Python, PyTorch, RAY — May 2024 - August 2024

- Developed **an action advising framework**, Gen2Spec, that distills knowledge of generalist agents to train specialist agents in a continual learning setting. This framework accelerates the training of an RL agent via action advice from a transformer-based meta-RL agent that can adapt to unseen tasks.

Eatron Technologies — Istanbul, Turkey
Engineering Intern (all work is under NDA) || Python, C++, PyTorch, ROS — June 2020 - June 2021

- Designed **graph convolutional** and **self-attention**-based neural network architectures to extract spatio-temporal features of a traffic scene for trajectory prediction in a Level-2+ ADAS powered vehicle.

ROKETSAN (in collaboration with Bilkent University) — Ankara, Turkey
Industrial Design Project (all work is under NDA) || Python, C++, PyTorch, ROS — Sept 2019 - June 2020

- Developed a mobile robot that follows a human leader, combining **YOLOv3 for object detection** and **artificial potential field method for path planning** in a mapped area with unknown dynamic obstacles.

Research Work

Scaling Curriculum Learning For Autonomous Driving — *ArXiv, 2026*
***Cevahir Koprulu**, David Paz, Feng Tao, Yuliang Guo, Xinyu Huang, Ufuk Topcu, Liu Ren*

Neural Stochastic Differential Equations for Uncertainty-Aware, Offline Reinforcement Learning — *ICLR, 2025*
***Cevahir Koprulu**, Franck Djeumou, Ufuk Topcu*

Safety Prioritizing Curricula for Constrained Reinforcement Learning — *ICLR, 2025*
***Cevahir Koprulu**, Thiago D. Simão, Nils Jansen, Ufuk Topcu*

Dense Dynamics-Aware Reward Synthesis: Integrating Prior Experience with Demonstrations — *L4DC, 2025*
***Cevahir Koprulu**, Po-han Li, Tianyu Qiu, Ruihan Zhao, Tyler Westenbroek, David Fridovich-Keil, Sandeep P. Chinchali, Ufuk Topcu*

Joint Learning of Reward Machines and Policies in Environments with Partially Known Semantics — *Artificial Intelligence, 2024*
*Christos Verginis, **Cevahir Koprulu**, Sandeep Chinchali, Ufuk Topcu*

Risk-Aware Curriculum Generation for Heavy-tailed Task Distributions — *UAI, 2023*
***Cevahir Koprulu**, Thiago D. Simão, Nils Jansen, Ufuk Topcu*

Reward-Machine-Guided, Self-Paced Reinforcement Learning — *UAI, 2023*
***Cevahir Koprulu**, Ufuk Topcu*

Act to Reason: A Dynamic Game Theoretical Driving Model for Highway Merging Applications — *CCTA, 2021*
***Cevahir Koprulu**, Yildiray Yildiz*

Technical Skills

- Programming languages: (Expert) Python, (Proficient) C++, Julia
- Frameworks: PyTorch, JAX, TensorFlow, ROS, RAY, verl, vLLM, Hugging Face Transformers

Relevant Coursework

- **UT Austin**: Online Learning, Causality and Reinforcement Learning, Statistical Machine Learning, Learning-based Optimal Control, Game-Theoretic Modeling of Multi-Agent Systems, Program Synthesis, Cyber-Physical Systems, Reinforcement Learning, Convex Optimization, Probability and Statistics, and Complex Networks in the Real World.
- **École polytechnique fédérale de Lausanne** (Exchange - Spring 2019): Image Analysis and Pattern Recognition, Convex Optimization, Deep Learning, and Biological Modelling of Neural Networks.
- **Bilkent University**: Statistical Learning and Data Analytics, Robust Feedback Theory, Introduction to Financial Mathematics, Neural Networks, and Game Theory.

Research and Leadership Experiences

Reinforcement Learning Reading Group — Austin, TX
Organizer — Aug 2025 - Ongoing

- RLRG at the CS Department of UT Austin is a student-run group that discusses research papers on RL.

Center for Autonomy — Austin, TX
Ph.D. Student — Aug 2021 - Ongoing

Systems Laboratory — Ankara, Turkey
Undergraduate Researcher — Sept 2019 - July 2021

- Developed **human driver models** from real-traffic data that can change its reasoning level dynamically by combining **level-k game theory** and **reinforcement learning**.

IEEE Robotics and Automation Society at Bilkent University — Ankara, Turkey
Chairman — May 2017 - June 2018

- Organized "Mühendis Kafası" in cooperation with Technology Development Foundation of Turkey: Series of sessions on Computer Vision and Deep Learning.
- Gave lectures on robotics, control techniques, and related micro-controller programming: EE-101: Introduction to Robotics with Arduino.

Language Skills

- Turkish: Native proficiency
- English: TOEFL 110/120 (Fall 2020)
- French: DELF B1 (Spring 2015)

Recreational Interests

I enjoy climbing, cycling, watching/playing soccer (football :)), and learning about history and psychology.